CITS4403 Project: Modelling the Perth Housing Crisis

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1 Background

Housing instability currently exists as a critical socioeconomic issue within Australian society. Despite high levels of economic prosperity, escalating housing pressures and persistent homelessness continue to affect Australians [3, 4], placing disproportionate burden on marginalised and lower-income groups, such as young people, migrants and renters [7].

Over the past two decades, housing and rental costs in Australia have increased significantly relative to incomes, with home-ownership rates for all age groups under 65 decreasing, particularly among the younger demographic and lower income households [7]. In addition to these general affordability issues, multiple interest rate increases by the Reserve Bank of Australia (RBA) have heavily increased mortgage repayments, creating more financial stress for those lucky enough to enter the housing market [6].

Our model works to act as a controlled environment allowing the discovery and analysis of the factors driving housing instability in Perth. While the majority of studies regarding the housing market are typically based within an Australian context, our model is specific to landscape of the Perth metropolitan region.

2 Research Hypothesis

We aim to determine which factors have the greatest impact on home ownership rates and housing instability in Perth. Specifically, how factors such as the recent interest rate hikes, the upfront deposit amounts required, and the general income inequality between agents affect housing accessibility. Ultimately, we plan to use our model to evaluate these variables to discover which factor has the most significant influence on decreasing Perth's current housing instability.

3 Prior work and Contributions

Many prior works attempt to simulate housing markets across varying contexts and approaches. Housing crises in London [8] and Tokyo [17] are examined via the implementation of system dynamics models and agent based models respectively, with other works focusing on the design of targeted interventions for such scenarios [9]. Models developed in the Australian context often seek to support policy decisions. Economic modelling approaches are used in a 2025 paper to estimate housing need in the Australian market [9], whilst a 2020 model of the Australian housing market seeks to examine relationships between construction, vacancies, rents and prices, via an empirical modelling method, finding low interest rates and high immigration as key factors influencing rapid growth in housing prices and construction [16]. Simple agent based models simulate generic housing markets, with J M Applegate presenting an extendible agent based model, simulating interactions between households buying and selling houses of varying quality [5].

Whilst models exist for the simulation and examination of both housing crises, and the Australian market, we seek to combine these elements, with a specific focus on the Perth metropolitan region. Similarly to Saunder's 2020 paper [16], we aim to identify relationships involving interest rates, rates and prices, however aim to do so with specific focus on outcomes for renters. Uniquely, we aim to simulate the Perth market via an agent-based model, similar to that presented by Applegate, however with significant extensions and assumptions tailored to our context. Our approach allows us to simulate specific interactions between renters

and landlords, offering a visual representation of the distribution of household rental status and characteristics across the city, alongside renter specific outcomes.

We make the following contributions:

1. We present an **agent-based model** of the housing market in the Perth metropolitan region
2. Via simulation, we investigate the **impact on renter outcomes associated** with interest rate hikes, upfront deposit amounts required, and the general income inequality within Perth.

4 Model specification

4.1 Classes

`Dwellings`:
Represents the houses in our model. Each `Dwelling` is initialised with a tuple representing a set of coordinates (*coord*), which then determines it's *quality* based on its location to hotspots, such as the Perth CBD, Nedlands, Peppermint Grove, and Cottesloe, as visible in Figure 1, with higher quality areas shown in yellow and lower quality in blue. Dwellings also store the current *occupant* and *owner*, (None when initialised), alongside whether it is for sale or not (*isOnMarket*). Lastly, each `Dwelling` has a rent and sale price, formulated using the `AVERAGE_HOUSE_PRICE` (or `AVERAGE_RENT` if it's being used as a rental), multiplied by it's quality score, and then further multiplied by a random factor to account for real world variations.

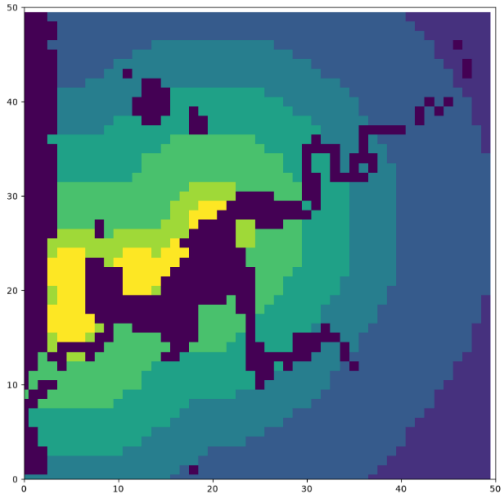


Figure 1: Dwelling quality across the map.

`Household`: Represents the people occupying the houses in our model. They're initialised with an income using a lognormal distribution, with the mean being the `AVERAGE_INCOME` variable, and their cash assets (*money*) are set to double their income, so the household can start buying/renting houses immediately (if applicable). A `Household` can occupy a house (*occupying*), and can own multiple houses and mortgages (*owns* and *mortgages* respectively).

`Bank`: Represents the bank. Each `Bank` is initialised with a base interest rate (*interestRate*), and tracks all the active mortgages (*activeMortgages*) occuring in the model. It also hosts an method to update the interest rate each year to simulate the rate changes, with the interest rate unable to go above 17.5% or below 0.1% based on historical data [15].

Mortgages: Holds information about each households' mortgage (or mortgages). It tracks the household who is responsible for paying off the loan (*borrower*), and the property that is tied to the mortgage (*dwelling*). It also holds the *initialLoanAmount* for the loan, along with the year's *interestRate*, how many years the loan has been active for (*yearsPaid*), and the remaining balance on the loan (*loanAmount*). The `Mortgages` class also has a method to calculate the annual payments a household has to make towards their mortgage by using the annuity formula, which most Australian banks implement [20][11].

4.2 Tick() Method

The `tick()` method in our model represents a year passing, in which households can buy, sell or rent out houses.

1. The bank updates the index rate, which also serves as the interest rate. This in turn updates all the house prices to match the new rate.

2. A random selection of 10% of the households leave the simulation, allocating their houses back to the market and clearing their loans. New households will take their place.

3. Households then get paid their yearly income and make mortgage and rent payments (if applicable). If they can't afford the mortgage, the bank takes the house.

4. Houses that are on the market that haven't been sold have their prices lowered. Households with investment properties list a houses for sale if its price is 150% greater than the price they brought it for.

5. Households can purchase a house, and are given 9 chances with 80% chance to be allowed to purchase an additional house.

6. If unable to own a house, households can then try and rent a house.

4.3 Assumptions

Income: Median household income (weekly): $1,865. Income is modelled on a heavy tailed logonormal distribution, allowing the simulation of many agents with similar income, but few agents with significantly higher incomes. [2].
Income Inequality: The logonormal distribution's `sigma` value of 0.55 results in a gini coefficient of 0.307, mirroring Australia's gini coefficient in 2022-3 [1].
Housing Prices: Median price data: $965,000 [14].
Rental Prices: Based on the median rental prices (weekly): $750 [14].
Interest Rate: Based on the cash rate implemented on the 30th of Sept 2026 [15]: 4.6%.
Profit Margin: Assuming that home investors will sell their property once it is valued at 150% its original cost, with the 150% figure chosen arbitrarily.
Average Loan Period: Based on the average loan period for most mortgages [10]: 30 years.
House Deposit: Based on the average home deposit in Australia [19]: 20%.

5 Methodology

To investigate the factors influencing housing instability in Perth, we used an agent-based modeling approach, allowing us to simulate the interactions of households within the housing market, and how those interactions create patterns in the rental, home ownership, and homelessness rates. We also tracked the percentage of renters in step 10 that then go on to own a home by step 30 to see if households could escape the "rental trap", naming this percentage `Mobility`.

Firstly, we created a control simulation, with the **income inequality** variable set to 0.55, **deposit** set to 0.2, and the **interest rate** set to 0.046. We then ran this control multiple times in order to calculate the mean for each percentage rate (Renters, Owners, Homeless, Mobility), since a significant portion of our calculations involve randomness.

Then we changed the income equality to 0.40, 0.50, 0.60, 0.65, 0.70, 0.75 and 1.0, where the higher the number, the larger the gap between households with maximum income vs the minimum. When running these simulations, we kept the deposit and interest rate as it was in the control group. Again, we ran this multiple times to calculate the mean rates.

We then did the above for the deposit (0.05, 0.10, 0.15, 0.25 and 0.30) and interest rate variables (2.6%, 3.6% and 5.6%), only changing one variable at a time and keeping the rest the same as per the control group.

6 Results

Results are presented in Tables 1, 2 and 3. Generally, we can observe the proportion of renters increasing as income inequality rises, however no significant changes are visible for interest rate or deposit amount increases. Geographically, as can be seen in Figure 2, regardless of inequality, rate or deposit changes, higher quality areas consist of either owner occupied dwellings or unoccupied dwellings, with middling quality zones being populated mostly by renters, and lower quality areas being lived in by renters or uninhabited.

Inequality at:	0.40	0.50	0.55 (Control)	0.60	0.65	0.70	0.75	1.0
% of Renters	40.899%	47.183%	53.593%	49.882%	51.805%	53.863%	50.577%	49.325%
% of Owners	49.848%	40.722%	35.172%	36.184%	30.558%	29.352%	30.192%	23.904%
% of Homeless	9.253%	12.095%	11.235%	13.934%	17.637%	16.785%	19.231%	26.771%
% of Mobility	18.328%	16.587%	17.570%	16.958%	21.026%	22.147%	21.849%	23.099%

Table 1: Household distribution under changing income inequality levels.

Deposit at:	0.05	0.10	0.20 (Control)	0.30	0.40
% of Renters	50.337%	53.214%	53.593%	45.336%	52.733%
% of Owners	35.501%	33.544%	35.172%	39.980%	34.194%
% of Homeless	14.162%	13.242%	11.235%	14.685%	13.074%
% of Mobility	17.135%	17.327%	17.570%	16.207%	20.232%

Table 2: Household distribution under changing deposit requirements.

Interest Rate at:	0.1%	0.2%	4.6% (Control)	6.0%	10.0%
% of Renters	46.770%	50.194%	53.593%	39.389%	47.149%
% of Owners	37.399%	35.130%	35.172%	48.119%	40.216%
% of Homeless	15.832%	14.676%	11.235%	12.492%	12.635%
% of Mobility	25.458%	21.932%	17.570%	14.888%	13.886%

Table 3: Household distribution under changing interest rates.

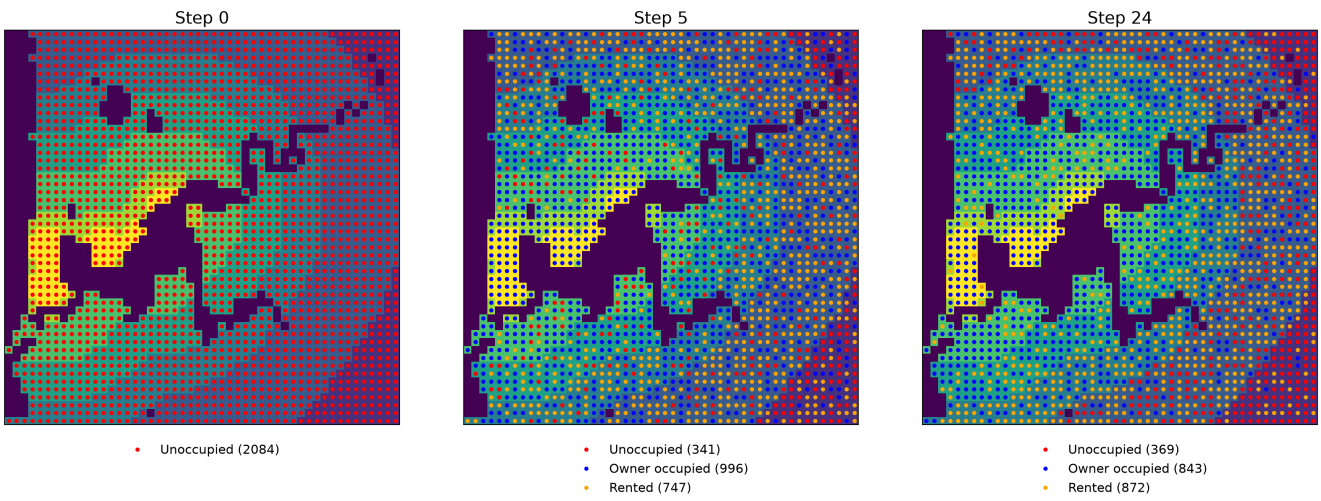


Figure 2: Household distribution over time, with $sigma = 0.55$, $deposit = 0.2$ and $rates = 0.046$

7 Discussion

7.1 Findings

Inequality Index: as the index increases in value (i.e. as the gap widens between the rich and the poor), the rate of homelessness steadily increases, peaking at 26.771% at an income inequality level of 1.0. In addition, the percentage of home owners notably decreases, aligning with our initial expectations. Whilst the rate of renters slightly fluctuates, mobility decreases as inequality is smaller than the control group and increases as the inequality level is larger than the control, showing that early renters are still managing to enter the property market.

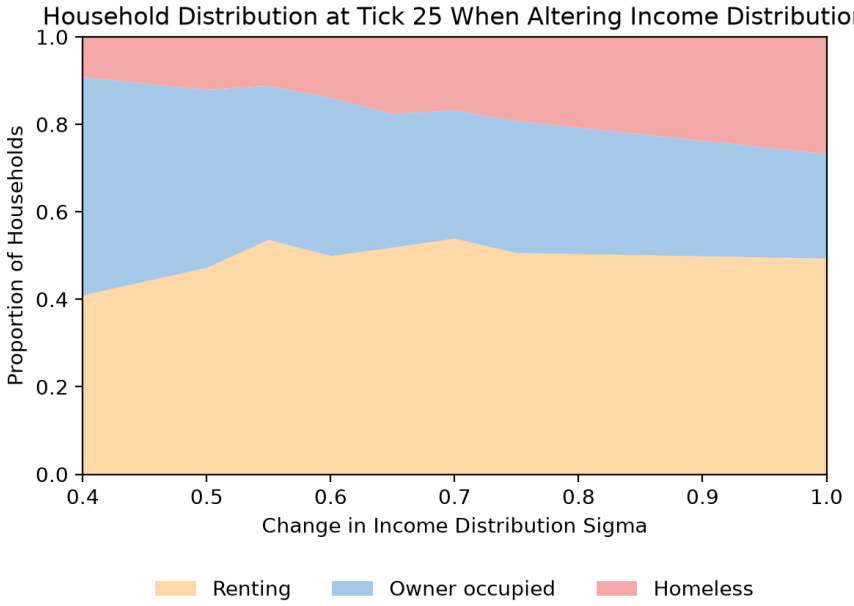


Figure 3: Household distribution when altering income inequality

Deposit Amount: as the deposit amount increases, it hypothetically becomes harder to obtain a mortgage due to the additional savings required. Interestingly, home ownership rates do not uniformly decline, hovering around the 35% mark. Homelessness rates also remain steady, not showing any major changes as the deposit amount required for a loan increases. However, the rate of renters stays consistantly high, representing approximately 50% of the households in the simulation. In addition, the extra cost doesn't deter renters from entering the property market, as when deposit = 0.4, mobility peaks at 20.232%.

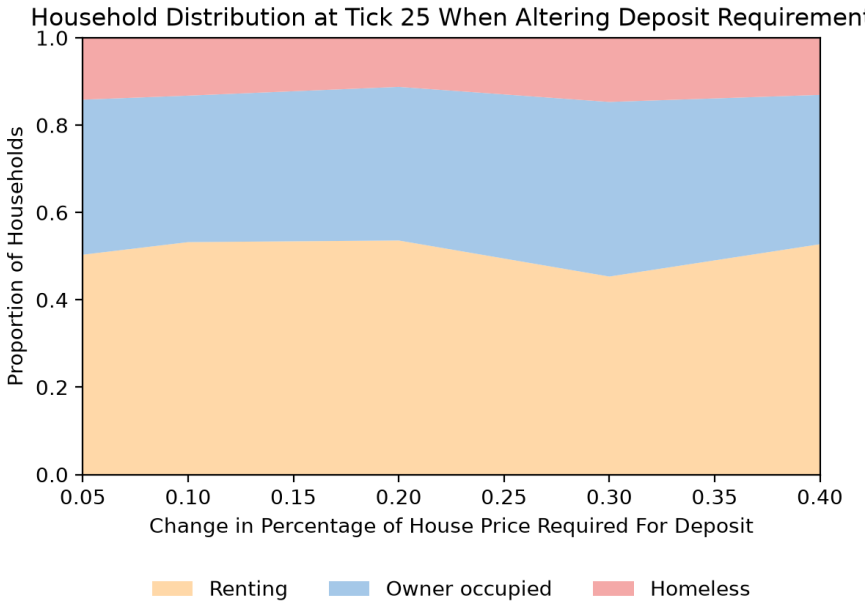


Figure 4: Household distribution when altering deposit requirements

Interest Rates: as the interest rate increased, we initially predicted that renters and homelessness would increase whilst the amount of home owners decreases, as mortgage repayments would have become more expensive. However, the results show nearly the opposite, with the percentage of home owners actually increasing as the rates rise (reaching a high of 48.119% when rates are 6.0%), On the other hand, the percentage of renters drops to 39.389% at the same level. Homelessness is actually the highest when the interest rate is at its lowest (15.832% when rates are 0.1%) which we did not predict. However, mobility shows a consistent decline as rates rise, falling from 25.458% to 13.886%, which is more in line with what we expected.

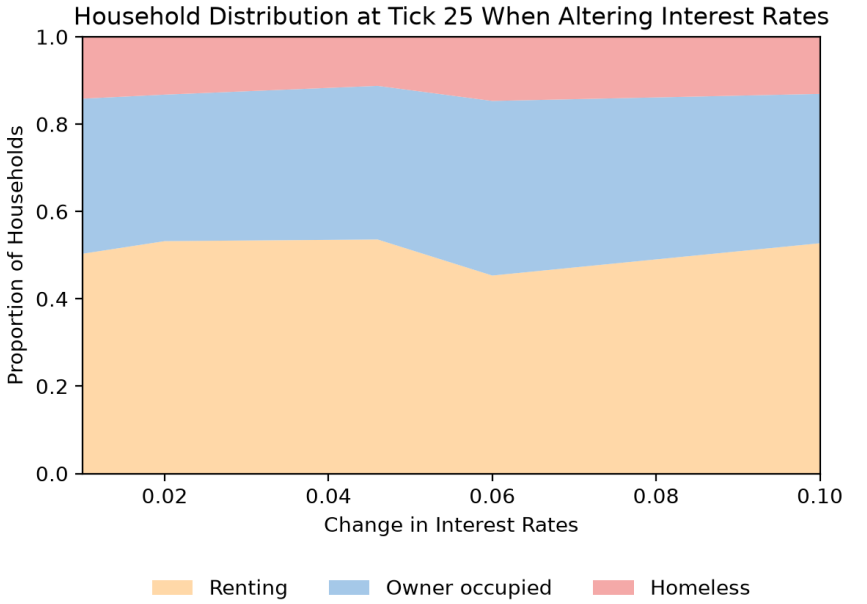


Figure 5: Household distribution when altering interest rates

7.2 Limitations

Limitations exist within our model, including:

- The formula used to update the interest rate occurs yearly in our model, whereas in reality it can be updated potentially up to 8 times a year based upon the RBA cash rate board meetings, with the rates being increased four times so far in 2026 [18].
- Our formula to calculate mortgage payments is based on interest compounding yearly, and households also making one annual payment. In reality, most interest rates compound monthly with households making monthly or fortnightly payments.
- In our simulation, we only have one dwelling type (houses). In reality, there are many different dwelling types, such as apartments, townhouses, and duplexes.
- Lower deposit amounts are typically saved for first home buyers, which our model doesn't track.
- In addition, when a household takes out a loan with less than 20% deposit saved, they also normally have to pay for Lenders Mortgage Insurance (LMI)[13], which isn't included in our model.
- Our model only considers standard mortgages from banks, and does not consider other loans such as having a guarantor, or having another household buy a house for you (such as the 'Bank of Mum and Dad').
- Our model does not take into account factors such as comparing different banks for loans, building inspections, settlement and stamp duty costs, and home insurance to name a few [12].

8 Conclusion

Based on our model's findings, we can address the hypothesis by evaluating how each factor influences home ownership and housing instability in Perth. The results indicate that income inequality has the largest impact on housing accessibility. When the income inequality gap is widened, it makes it harder for households to find houses in their price range and afford loan repayments, therefore resulting in lower ownership and higher renter and homelessness rates. Deposit requirements had a less predictable effect than anticipated. Rather than falling as larger deposits were required, home ownership rates only slightly varied randomly, suggesting that households had possible adjusting their savings when faced with more costs to enter the housing market.

Interest rates displayed contradictory results to what we predicted, with home ownership rates increasing alongside the rates. This suggests that interest rates in isolation play a less dominant role in determining housing instability in our model. However, there is also the possibility that our model fails to capture the complexities associated with the influence of interest rates on housing instability, preventing it from revealing the relationship adequately.

In conclusion, based on our model's findings, income inequality has the most impact on Perth's housing market. In order to help bridge the gap, governmental interventions should focus on reducing wealth disparity and easing upfront cost barriers, with a targeted focus on helping low-socioeconomic households.

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